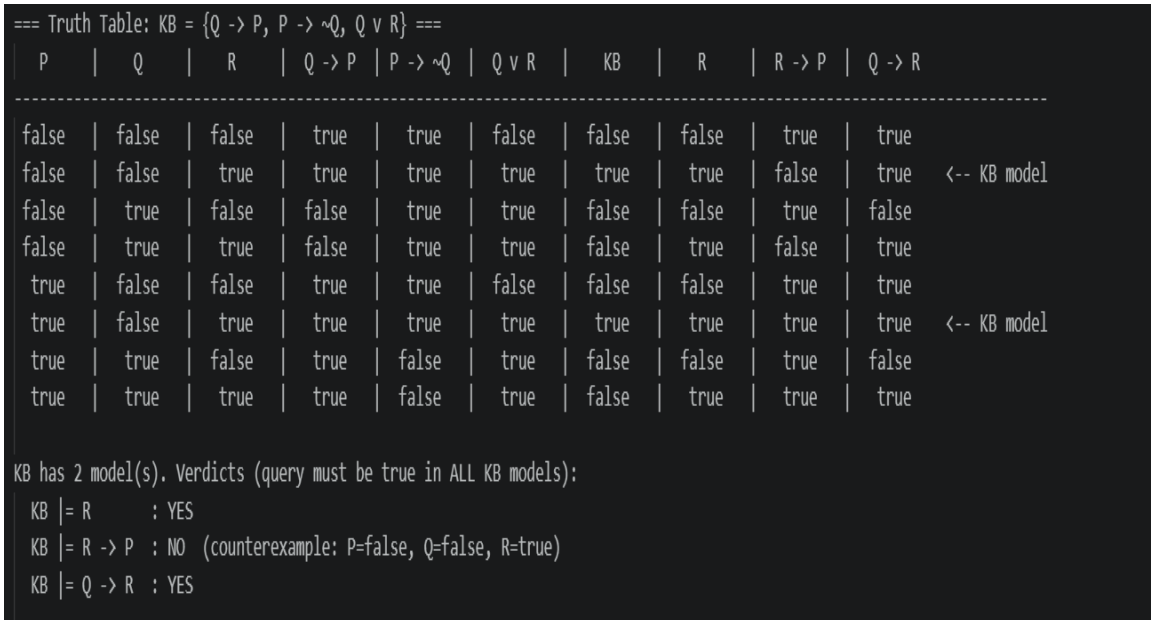
1BM24CS315-WEEK-06

1. Implement Propositional logic algorithm and show that given queries entails the Kb or not.

Output:



```
=== Truth Table: KB = {Q -> P, P -> ~Q, Q v R} ===
   P    |    Q    |    R    | Q -> P  | P -> ~Q  | Q v R  |    KB    |    R    | R -> P  | Q -> R
-------------------------------------------------------------------------------------------------------------
 false  |  false  |  false  |  true   |  true    |  false |  false   |  false  |  true   |  true
 false  |  false  |  true   |  true   |  true    |  true  |  true    |  true   |  false  |  true    <-- KB model
 false  |  true   |  false  |  false  |  true    |  true  |  false   |  false  |  true   |  false
 false  |  true   |  true   |  false  |  true    |  true  |  false   |  true   |  false  |  true
  true  |  false  |  false  |  true   |  true    |  false |  false   |  false  |  true   |  true
  true  |  false  |  true   |  true   |  true    |  true  |  true    |  true   |  true   |  true    <-- KB model
  true  |  true   |  false  |  true   |  false   |  true  |  false   |  false  |  true   |  false
  true  |  true   |  true   |  true   |  false   |  true  |  false   |  true   |  true   |  true

KB has 2 model(s). Verdicts (query must be true in ALL KB models):
  KB |= R       : YES
  KB |= R -> P  : NO  (counterexample: P=false, Q=false, R=true)
  KB |= Q -> R  : YES
```